

# Multi-Agent Financial Risk Assessment System

Simulate financial scenarios, adjust assumptions, and observe impact on risk metrics.





# Project Overview & Objective

## Motivation

Scenario-based financial risk assessment for robust decision-making.

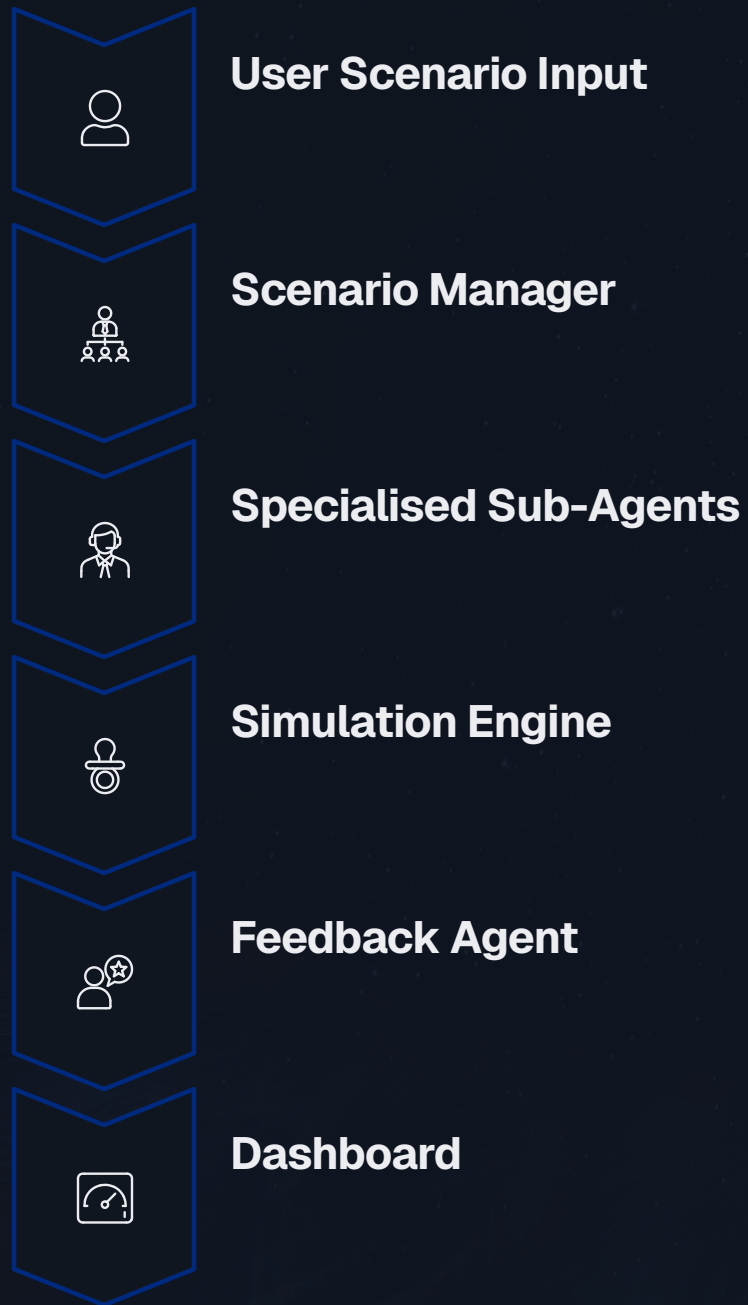
## Goal

Simulate "what-if" scenarios (e.g., interest rate shocks, FX movements) and observe their impact on risk metrics.

## Emphasis

Scenario-driven analysis, deterministic financial logic, and explainable outputs.

# System Architecture: High-Level Pipeline



Agents communicate via a **Scenario ID** and **Scenario Date**.

# Data Layer: PostgreSQL Database

## Master Table: financial\_statements\_monthly

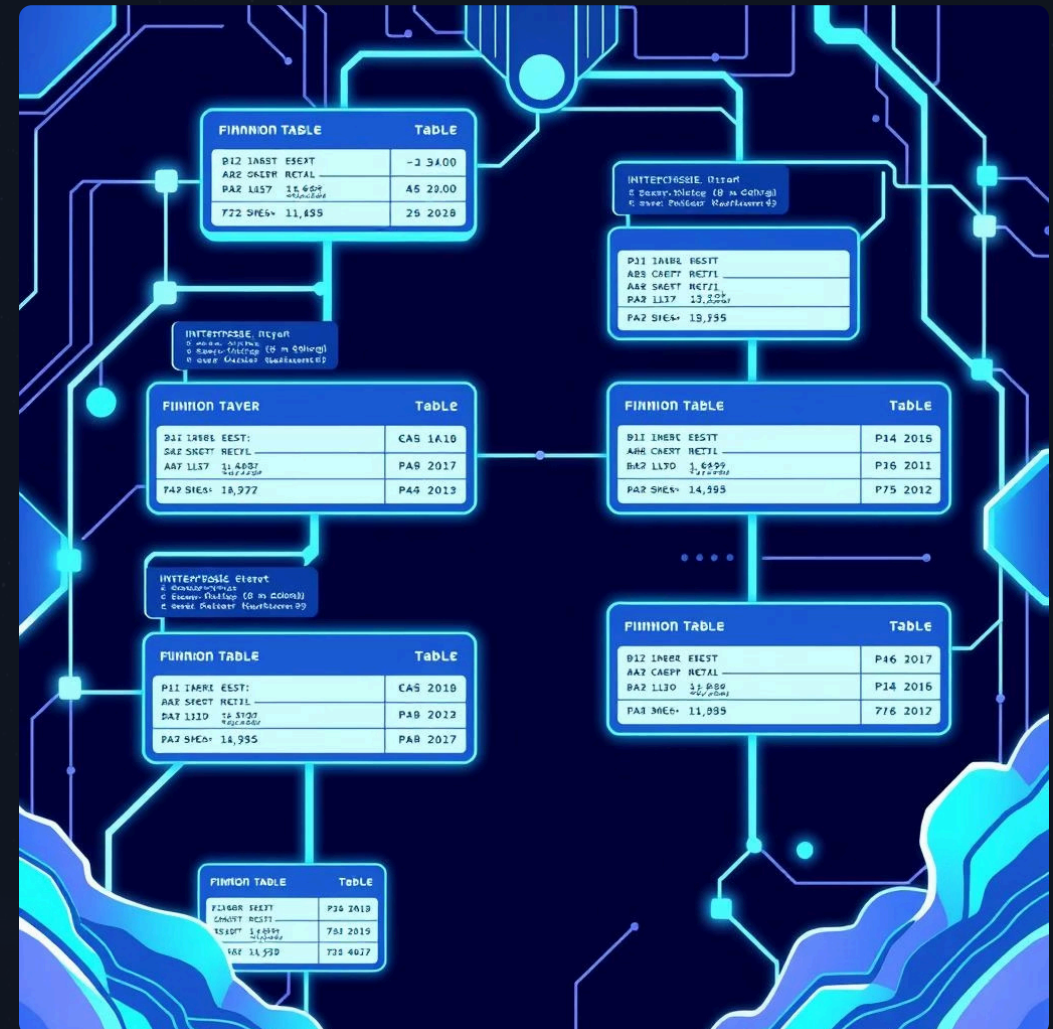
- Monthly granularity (2020–2024)
- Single source of truth (Income Statement, Balance Sheet, Cash Flow, Treasury data)
- Wide table for efficient scenario simulations

## Support Table: debt\_instruments

- Loan-level granularity
- Captures debt structure, currency exposure, maturity risk
- Used by Debt Agent for analysis

## External Table: market\_indices\_daily

- Daily macroeconomic data
- Includes FX rates, interest rates, risk-free rate, inflation



# Scenario Manager: The Orchestrator

The Scenario Manager acts as the central orchestrator, leveraging an **LLM** to interpret user scenarios.

- Interprets user scenario descriptions
- Identifies affected financial domains
- Asks for essential parameters only

📄 **Key Principle:** Unspecified parameters default to historical values, preventing user overload and ensuring realistic, deterministic scenarios.



# Specialised Sub-Agents



## Liquidity Agent

Focuses on short-term survival and operational liquidity. Metrics include Cash Conversion Cycle, Burn Rate, and Current Ratio.



## Debt Agent

Assesses solvency, leverage, and financing risk. Metrics include DSCR, Debt/EBITDA, and Interest Sensitivity.



## Market Agent

Monitors external macro and FX risk. Metrics include FX Net Position, FX VaR, and Degree of Operating Leverage.

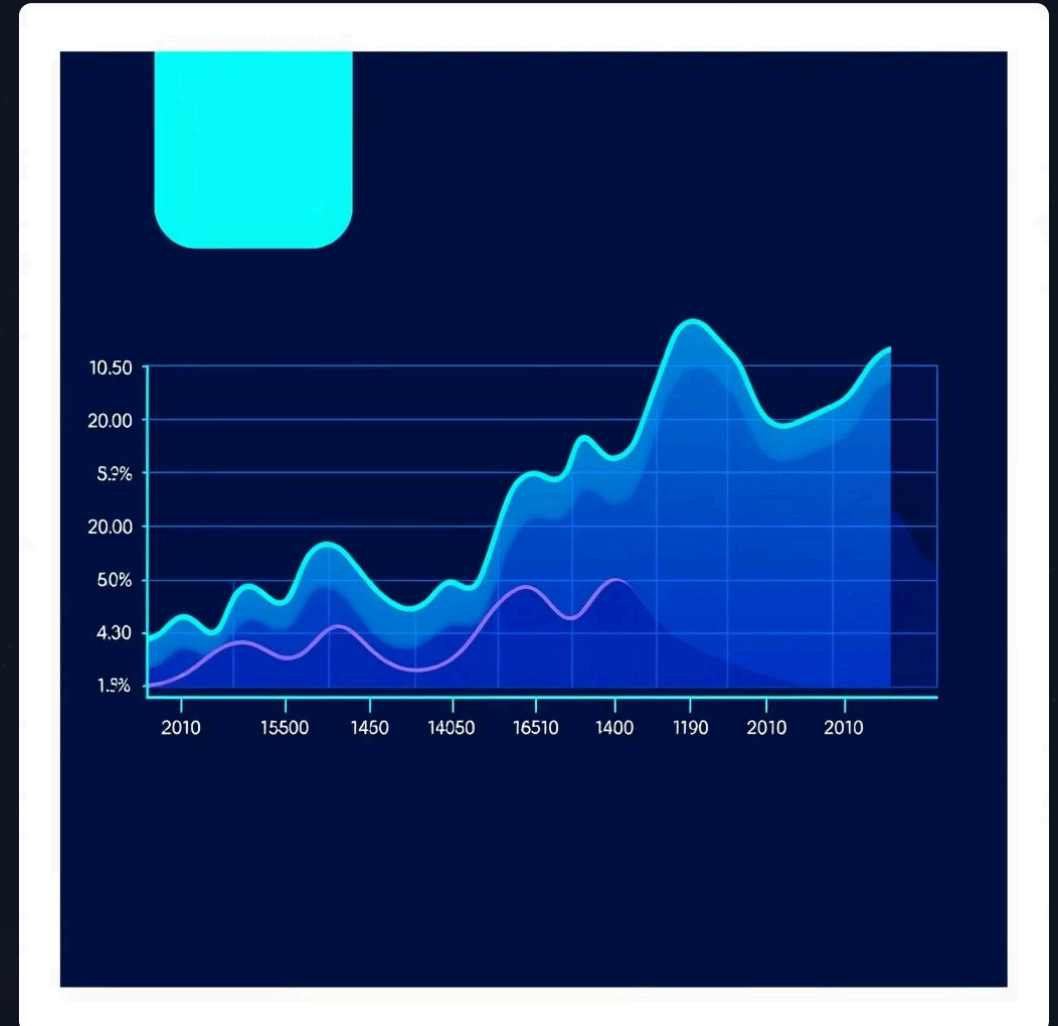
# Scenario Logic: Historical Data + Parameters

Historical financial data forms the **baseline** for all simulations.

Scenario parameters apply **controlled shocks or adjustments** to this baseline.

- Same equations are used
- Different inputs lead to different outcomes per scenario

This approach ensures **comparability, explainability, and auditability** of results.



# Simulation Engine: Monte Carlo

The Monte Carlo simulation engine quantifies uncertainty and stresses extreme outcomes.



## Distribution-Based

Metrics are treated as distributions, not single values.



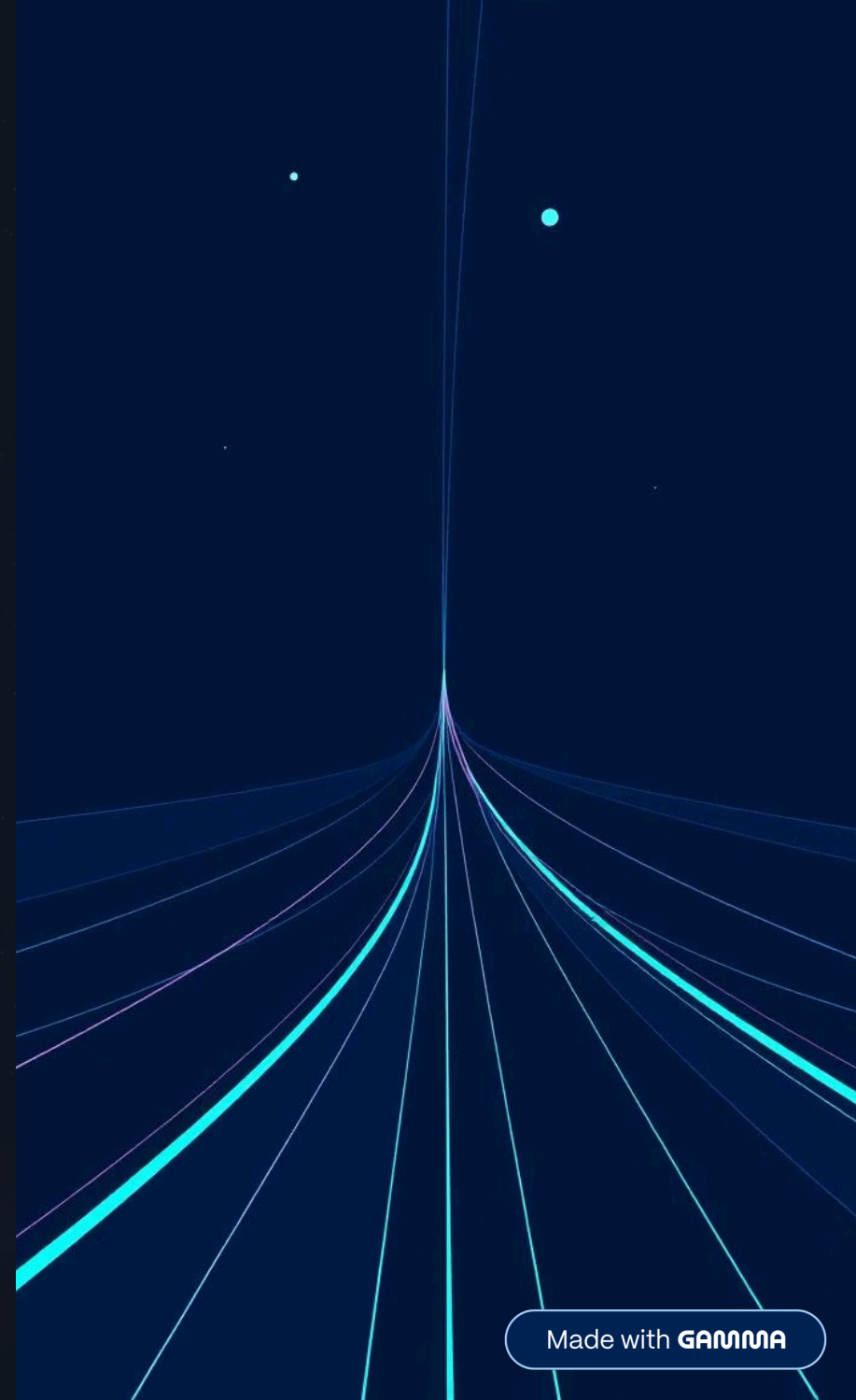
## Thousands of Paths

Generates thousands of simulated paths to explore potential outcomes.



## Risk Quantification

Estimates risk ranges and quantifies uncertainty in financial projections.



# Feedback Agent & Dashboard

## Feedback Agent

- Interprets simulation results
- Detects risk threshold breaches
- Generates natural-language insights and warnings

*Example: "DSCR falls below 1.2 in 35% of scenarios."*

This acts as the **explainability and decision-support layer**.

## Dashboard & Visualisation

- Visualises scenario comparisons
- Displays risk metric distributions
- Provides time-series stress views

Offers executive-friendly outputs with drill-down capability.



# Key Design Principles & Conclusion

1

## **Deterministic Logic**

Ensures consistent and auditable financial calculations.

2

## **Minimal User Input**

Streamlines the process, focusing on essential parameters.

3

## **Agent Specialisation**

Leverages dedicated agents for specific financial domains.

4

## **Explainable AI**

Provides clear insights and warnings for informed decisions.

5

## **Scalable Architecture**

Designed for growth and adaptability to evolving financial needs.

This multi-agent system offers superior, scenario-driven risk analysis, supporting strategic financial decision-making.

# Thank You